



Research Paper

Implementation and validation of optimal start control strategy for air conditioners and heat pumps

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ABSTRACT

Commercial buildings are responsible for approximately 20 % of the total energy consumption and greenhouse gas emissions in the United States. Over 85 % of these buildings lack building automation systems, and many are small (<50,000 square feet), underserved, and use rooftop units (RTUs) for heating, ventilation, and air-conditioning needs. Because these buildings lack proper energy management systems, several operational deficiencies lead to excess energy consumption. Studies have shown that managing the RTUs' heating and cooling set points, schedules, setbacks, and optimal start can result in a 20 % to 25 % reduction in electricity consumption in small commercial buildings. These buildings typically use fixed schedules to start the RTUs 60 to 120 min before occupancy begins, which results in excess energy consumption. This paper presents research that demonstrates and evaluates the performance of four optimal start methods, which utilize data-based modeling as a key element in facilitating adaptive control in response to time-varying inputs while requiring minimal sensor inputs. The evaluation found energy savings in two commercial buildings equipped with RTUs by periodically alternating four different optimal start models during the cooling and heating season. The resulting energy savings are positive for all models and range from 2 to 5 kWh/day/unit. The units on the east side of the building showed higher savings, while interior units showed greater variability in savings due to the differences in capacities and room sizes. Savings were considerably greater during the heating season compared to the cooling season. The performance of all four models on Mondays was poor; models suggested a shorter optimal start time, which resulted in relatively larger errors. The future work will look at using a different model for the days after weekends and holidays.

1. Introduction

In the United States, heating, ventilation, and air conditioning (HVAC) systems account for 40 % of residential energy use and 30 % of commercial building energy use [1]. Several studies have highlighted common control issues in commercial buildings, including longer-than-necessary operation of HVAC and lighting systems [2], poor ventilation control during warm-up [3], malfunction of control algorithms [4], misuse of exhaust fans, and incorrect sensor installation locations [5]. Other studies have demonstrated that scalable and cost-effective intelligent building systems can result in energy savings greater than 30 % [6]. Intelligent building operations include supervisory controls optimized to respond to time-varying inputs (e.g., weather, occupancy, utility rates). However, small- and medium-size commercial buildings (SMBs) are less likely to employ a building automation system (BAS) and typically have fewer degrees of freedom for intelligent building

operation. These applications typically employ multiple packaged rooftop units (RTUs) controlled by individual thermostats. Therefore, it would be practical to develop an optimal start controller and embed it within a smart thermostat. Existing thermostats provide direct on/off control of RTUs and could be readily replaced with smart thermostats at relatively low cost.

Proper scheduling represents one of the best opportunities to minimize energy costs in SMBs [7,8]. Many SMBs use programmable thermostats to automatically shut off RTUs based on time-of-day and day-of-week scheduling. Indoor temperatures are generally set back at night when buildings are unoccupied to reduce energy use. It is common for the RTUs to be started before the beginning of occupancy and to operate long enough (>1h) for the indoor temperature to reach the desired occupied set point by the time people occupy the building. The time-of-day schedule is often programmed to start the system early enough to ensure that the building warms up or cools down fast enough on the coldest or warmest mornings in many buildings [9,10]. As a result,

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Nomenclature

α	parameter for optimal start model
σ	dead band
τ_r	time constant of the zone
C_r	thermal mass of the zone
e_{time}	time-based error
e_{temp}	temperature-based error
k	day
Q_{RTU}	the rated RTU cooling/heating capacity
R_O	thermal resistance between the exterior and the interior of the wall
R_W	thermal resistance between the interior wall and the zone
RQ_{RTU}	the product of RTU power and resistance
t	time
t_{act}	actual operation time
t_{opt}	optimal start time
T_o	outdoor air temperature
T_z	zone temperature
T_W	wall temperature
$T_{o,r}$	reference outdoor air temperature
T_{sp}	indoor temperature at occupancy start time
u	control signal

however, on all other days of the year, the system starts much earlier than necessary. This increases the number of operating hours and potentially increases energy use up to 5 kWh/day/unit.

An alternative approach to minimize the excess energy consumption is to use optimal start control, which reduces the equipment runtime without compromising occupant comfort. For simplicity, many building operators define a fixed operational schedule for the entire building. However, if different areas of the building have significantly different occupancy patterns, this approach wastes energy. The entire building may operate to maintain occupied temperature set points even though only some zones are in use. The optimal start control creates separate operating schedules for building zones with different occupancy patterns. An accurate optimal start algorithm uses real-time measurements (zone temperature, outdoor air temperature, etc.) to determine the necessary preconditioning time for each zone. The controller then delays starting the RTU until the latest possible moment, ensuring that each zone reaches the desired temperature just in time for occupancy.

Existing optimal start methods are based on research that is several decades old. One study [11] developed a rule-based optimal start control with a predictive occupancy inference method for radiant floor heating. By implementing the optimal start-stop control in a simulation environment, heating energy consumption was reduced by up to 3 % compared to the time-of-day schedule control. In another study [12], a rule-based strategy is used with a zone thermal model that can account for several factors (e.g., thermal properties, HVAC system, weather conditions) in residential buildings. Another study [13] developed an optimization framework based on gray box modeling. The input data used in the gray box model included operating data, and the heat transfer rate was predicted using a linear regression model. For optimal cooling, they used a shortest path-based heuristic algorithm to solve the optimization problem. Another study [14] developed a control strategy to optimize cooling supply among individual zones as well as chiller operation. However, the main drawback of typical optimal start controls is that once the model has been built, it remains fixed for the duration of the operation. There is a perceived risk that those control strategies may not work or may disrupt operations without capturing the thermal behavior of building in response to environment conditions.

Over the past decades, several papers have predicted the thermal

behavior of a building under varying external conditions. Bilous et al. [15] developed a practical tool for predicting the building heating loads based on a multivariate regression model for zone temperature prediction and a variety of internal variables and climate factors. Detailed hourly energy simulation software, EnergyPlus, and the prediction model were combined to analyze the time-varying factors influence on the thermal behavior of a building. Andelković and Bajatović [16] proposed the data-driven model of hourly natural gas consumption prediction based on the weather forecast data. The proposed method integrates weather forecasts and artificial intelligence to enhance prediction accuracy. Cholewa et al. [17] proposed a simplified short-term forecasting model that accounts for the actual heat power supply and outdoor weather conditions (i.e., temperature, wind speed, and solar insolation) of a building for predicting the heating energy consumption. The method is based on multiple linear regression using monitored data delivered to the heating system. The method was updated to consider occupant behavior as an input variable, enhancing prediction performance. It also integrated a model predictive control (MPC) system for building heating operations, based on weather forecasts. The authors found that it is important to update the data profiles because the occupant behavior could change over time. Many recent studies used time-varying inputs (e.g., weather, occupancy, utility rates) in the models. However, SMBs typically do not have sensing and environmental monitoring infrastructure (i.e., humidity, wind speed, and solar insolation). Also, because it is difficult to directly measure the building operation patterns, the thermal behavior of the building is estimated based on variables related to known characteristics of the building and the outside temperature.

The optimal start strategies have recently been replaced by data-driven techniques that use models of response times and parameter estimation methods to fit the models to observed data. Various model forms have been proposed, including numerous regression-type models, artificial neural networks (ANN), and MPC. Yang et al. [18] proposed the optimal start time strategy based on an optimized ANN model to determine the optimal start time for a heating system in a building. The zone thermal model uses the finite difference method, and the ANN model was trained using simulation data that contains room and outdoor air conditions. In another study [19], a two-time scale coordinated optimal control based on a stochastic scheduling scheme and a real-time optimal control scheme was developed to forecast load and generation uncertainty and identify the optimal start time for scheduling optimization. Additionally, Dai et al. [20] introduced a Q-learning-based control strategy to address optimization challenges in air-handling units (AHUs). By using a Q-learning agent, they were able to optimize the valve control for AHUs, each with differing performance. By opening the valves differently for each AHU, it was possible to reduce the precooling time by 2.48 (hour/month). Another study [21] proposed a hybrid MPC approach with a neural network-based feedback linearization technique for minimizing energy consumption in commercial HVAC systems. The optimal start-stop MPC achieved energy savings of up to 41 % compared to the conventional baseline control strategy in the simulation environment. In another study [8], an integrated HVAC management and optimal start scheduling algorithm was developed for peak load reduction. They reduced peak demand by conducting optimal set point control using MPC and combining it with a schedule control algorithm using mixed integer linear programming. A predictive control strategy for intermittent heating in office buildings was developed in another study [22] to improve energy efficiency and maintain indoor comfort. They used a physical model based on the state space method to simulate and forecast the building's thermal behavior. The simulations showed energy savings of up to 20 % compared to conventional control strategies.

In general, the start time of RTUs can be optimized while considering occupant thermal comfort based on a comprehensive knowledge of several factors such as the outdoor environment, the thermal behavior of the building envelope, and the thermal dynamics of RTUs. It is often challenging when designing the optimal start algorithm because of a

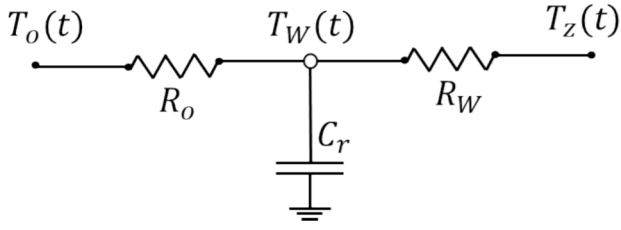


Fig. 1. 2R1C model.

lack of information about the characteristics of the building and RTUs. For example, the main drawback of MPC is that it requires an accurate model of the building and RTUs. The complexity of implementing the optimal start strategies is also a significant limitation because of the high implementation costs associated with providing site-specific solutions relative to the potential savings. Additional implementation costs are due to additional sensor requirements and labor to engineer and program the site-specific solutions. For this reason, the dynamics of modern RTU performance have not been closely evaluated around the optimal start control strategies. In addition, the high implementation time and cost made its commercial implementation impractical. Because of these issues, a data-based model enabling adaptive control in response to time varying inputs could be an appealing option for a conservative building manager because detailed building data such as wall thickness and material are not needed. The data-based model postulates a model structure and identifies the coefficients of that model from in situ data such as indoor temperature and set points to estimate the thermal characteristics of the building and the thermal performance of RTUs. Also, many nonlinear and time-varying effects take place, making the application of adaptive control techniques a reasonable choice. For these reasons, this paper presents and evaluates four different data-based adaptive optimal start control models.

The novelty of this research is to demonstrate and evaluate four optimal start methods that employ data-based modeling as key element in enabling adaptive control in response to time varying input with few sensor requirements. The paper presents data-driven optimal start strategies that are feasible in SMBs with RTUs and discusses the energy savings performance of each of the four methods using the results from the field test. The data-driven models estimate the time required to bring the indoor condition from the setback temperature to the occupied set point. The models' design reduces sensor and configuration requirements, making optimal start more cost-effective for SMBs.

Compared with the previously reported work, the contributions of this paper are threefold.

- 1) The data-driven adaptive optimal start controls, which respond to environmental variables, were developed, demonstrated, and evaluated. They calculate the minimum time needed to bring the zone temperature to the desired set point by occupancy, thereby saving energy while maintaining thermal comfort.
- 2) The paper also provides field validation results of the optimal start control of RTUs, which can be easily installed and integrated with existing SMBs. The proposed models can automatically estimate the necessary parameters for optimal start.
- 3) The energy savings were evaluated in two SMBs equipped with RTUs by periodically alternating the four different optimal start models during the cooling and heating seasons. The resulting energy savings are positive for all models and range from 2 to 5 kWh/day/unit.

In this paper, Section 2 describes the concepts and formulation of four different models and introduces parameter estimation methods. The optimal start controller's adaptive learning process relies on sensor data to identify/update the optimal start parameters without the need for the user interaction. Section 3 describes the systems and test conditions used to examine the four different optimal start strategies in two

commercial buildings. Section 4 describes the field performance of the optimal start methods including estimating the reduction in the energy costs. Conclusions from the study are documented in Section 5. The optimal start runtimes for all RTUs used in this analysis are included in the Appendix.

2. Methodology

This study explored four different data-driven optimal start control methods to determine the start time of RTUs. While the MPC approach requires a comprehensive knowledge of all the characteristics of the building and RTUs, the data-driven approach uses data sources (e.g., indoor, and outdoor temperature variations over time) to estimate the performance of a RTU, and leverage the information to predict the indoor temperature changes through RTU operation, ultimately determining the optimal start time. This section describes the four optimal start methods along with the parameter estimation techniques for these methods, validation approaches for accuracy assessment of these methods, and the method for estimating energy savings.

2.1. HYPERLINK "SPS:id::Sec2" Models to predict response time

This section provides a detailed description of each model used. Three main factors influencing the optimal start time are 1) the thermal characteristics of the zone, 2) the quantity of energy already stored within thermal mass in the zone, and 3) the thermal performance of the RTU. The models outlined in this paper are formulated using a heat transfer model based on thermal resistance (R) and capacitance (C), which has been widely adopted in zone modeling, as shown in Fig. 1.

The models are approximated as a 2R1C (two resistances and one capacitance) structure and are formulated with semi-empirical relationships between the optimal start time before occupancy and the measured temperatures. T_o is outdoor air temperature, T_z is the zone temperature, T_w is the wall temperature, R_o is the thermal resistance between the exterior and the interior of the wall, R_w is the thermal resistance between the interior wall and the zone, and C_r is the thermal mass of the zone located between two thermal resistances. The dynamics of indoor temperature T_z are governed by the following differential Eqs. (1) and (2):

$$\frac{dT_z}{dt} + \frac{T_z}{\tau_r} = \frac{Q_{RTU}}{C_r} + \frac{T_o}{C_r R_o} + \frac{T_w}{C_r R_w} \quad (1)$$

$$\tau_r = \frac{C_r R_w R_o}{R_w + R_o} \quad (2)$$

where τ_r denotes the time constant of the zone, Q_{RTU} represents the rated RTU cooling/heating capacity. Eq. (3) describes the zone temperature at time $t > 0$ after starting the RTU:

$$T_z(t) = RQ_{RTU} \left[1 - e^{-\frac{t}{\tau_r}} \right] + \beta [T_z(0) - T_o(0)] e^{-\frac{t}{\tau_r}} + \beta T_o(0) + [1 - \beta] T_z(0) \quad (3)$$

$$\beta = \frac{R_w(R_w + R_o)}{R_w(R_w + R_o) + R_o} \quad (4)$$

where RQ_{RTU} represents the product of RTU power and resistance $R (= R_w + R_o)$, which is the equivalent bulk thermal resistance through the wall. For ideal optimal control with startup at time $t = 0$, $T_z(t) = T_{sp}$ when $t = t_{opt}$, where T_{sp} is the desired indoor temperature (i.e., the occupied temperature set point) at occupancy start time. Hence, the optimal start time t_{opt} can be determined from Eq. (5):

$$t_{opt}(0) = \tau_r \ln \frac{RQ_{RTU} + \beta [T_z(0) - T_o(0)]}{RQ_{RTU} - T_{sp} + [1 - \beta] T_z(0) + \beta T_o(0)} \quad (5)$$

Eq. (5) is nonlinear and unsuitable for simple optimal control scheme. However, when the optimal start time is much shorter than the zone's time constant, Eq. (6) provides simplification:

$$e^{-t_{\text{opt}}/\tau_r} \approx 1 - t_{\text{opt}}/\tau_r \quad (6)$$

This simplifies Eq. (5) to:

$$t_{\text{opt}}(0) = \frac{\tau_r [T_{\text{sp}} - T_z(0)]}{RQ_{\text{RTU}} - \beta [T_z(0) - T_o(0)]} \quad (7)$$

Thus, Eq. (7) can be approximated as:

$$t_{\text{opt}}(0) = \frac{\tau_r}{RQ_{\text{RTU}}} [T_{\text{sp}} - T_z(0)] + \frac{\tau_r \beta}{RQ_{\text{RTU}}} [T_{\text{sp}} - T_z(0)] [T_z(0) - T_o(0)] \quad (8)$$

$$t_{\text{opt}}(0) = a_1 [T_{\text{sp}} - T_z(0)] + a_2 [T_{\text{sp}} - T_z(0)] [T_z(0) - T_o(0)] \quad (9)$$

Introducing a parameter a_3 to account for RTU startup time Eq. (9) becomes:

$$t_{\text{opt}}(0) = a_1 [T_{\text{sp}} - T_z(0)] + a_2 [T_{\text{sp}} - T_z(0)] [T_z(0) - T_o(0)] + a_3 \quad (10)$$

Rewriting this equation in discrete time form, the optimal time at the n^{th} sampling instant is given by Eq. (11):

$$t_{\text{opt}}(n) = a_1 [T_{\text{sp}} - T_z(n)] + a_2 [T_{\text{sp}} - T_z(n)] [T_z(n) - T_o(n)] + a_3 \quad (11)$$

Next, four different optimal start models are described, all of them derived from Eq. (11).

2.1.1. Model 1

The first model is formulated with two parameters assuming that the zone responds as if all thermal mass is concentrated in the indoor air, neglecting the influence of outdoor temperature. Model 1 can typically be used for interior zones or for well-insulated or high-mass buildings where the time required to reach operating conditions depends primarily on the indoor temperature at the beginning of the occupancy period. Rewriting Eq. (11) in discrete time form, the optimal start time can be estimated using Eq. (12). The optimal start time ($t_{1,\text{opt}}^k$) of current day (k) is determined based on two parameters ($\alpha_{1,a}^k$, $\alpha_{1,b}^k$), which are tuned from the previous days ($k-1$) data. The first parameter, $\alpha_{1,a}^k$ (minutes/1 degree²), denotes the rate at which the indoor temperature changes after the RTU starts conditioning the zone as shown in Eq. (13). The second parameter, $\alpha_{1,b}^k$ (minutes), denotes the amount of time required to change the indoor temperature by 1 degree as shown in Eq. (14).

$$t_{1,\text{opt}}^k = \alpha_{1,a}^k (T_{\text{sp}}^k - T_{z,0}^k)^2 + \alpha_{1,b}^k \quad (12)$$

$$\alpha_{1,a}^k = \frac{\Delta t^{k-1}}{(T_{\text{sp}}^{k-1} - T_{z,0}^{k-1})^2} \quad (13)$$

$$\alpha_{1,b}^k = t_{T_z^{k-1}=T_{z,0}^{k-1} \pm 1}^{k-1} - t_{T_z^{k-1}}^{k-1} \quad (14)$$

where Δt^{k-1} is the time, in minutes, required to change from the initial indoor temperature $T_{z,0}^{k-1}$ to the set point temperature T_{sp}^{k-1} after the RTU starts the day before ($k-1$), $t_{T_z^{k-1}=T_{z,0}^{k-1} \pm 1}^{k-1}$ is the time when T_z^{k-1} changes by 1 degree from $T_{z,0}^{k-1}$, while $t_{T_z^{k-1}}^{k-1}$ is the time when the RTU starts when the zone temperature is $T_{z,0}^{k-1}$.

2.1.2. Model 2

The second model determines the optimal start time ($t_{2,\text{opt}}^k$) using one parameter and the outdoor air temperature (T_o) adjustment, as shown in

Eq. (15). This model incorporates a reference outdoor air temperature ($T_{o,r}$) determined by geographical location, at which the RTU needs to operate continuously to maintain indoor thermal comfort. Based on $T_{o,r}$, this model determines the optimal start time by considering outdoor temperature variations between previous day (T_o^{k-1}) and current day (T_o^k). According to ASHRAE publications [23], $T_{o,r}$ is set to 0 °C (32 °F) in heating mode and 37.78 °C (100 °F) in cooling mode. Model 2 can typically be used in perimeter zones of a lightweight or poorly insulated building where the time required to reach operating conditions depends primarily on the outdoor air temperature at the beginning of the optimal start period. The parameter, $\alpha_{2,a}^k$ (temperature/hour), which is tuned from previous day ($k-1$) data, defines the rate of the indoor temperature changes from $T_{z,0}^{k-1}$ to T_{sp}^{k-1} over the RTU runtime to quantify how quickly the temperature changes with respect to time, as shown in Eq. (16).

$$t_{2,\text{opt}}^k = \frac{(T_{\text{sp}}^k - T_{z,0}^k)}{\alpha_{2,a}^k} \left(\frac{T_{o,r} - T_o^{k-1}}{T_{o,r} - T_o^k} \right) \quad (15)$$

$$\alpha_{2,a}^k = \frac{(T_{\text{sp}}^{k-1} - T_{z,0}^{k-1})}{\Delta t^{k-1}} \quad (16)$$

2.1.3. Model 3

The third model determines the optimal start time ($t_{3,\text{opt}}^k$) based on several factors, including the difference between the indoor temperature (T_z^k) and the occupied set point (T_{sp}^k), and the outdoor air temperature (T_o^k) with weather-compensated controls. This calculation is governed by Eq. (17) incorporating additional parameters:

$$t_{1,\text{opt}}^k = \alpha_{3,a}^k (T_{\text{sp}}^k - T_{z,0}^k) + \alpha_{3,b}^k (T_{\text{sp}}^k - T_{z,0}^k) \frac{(T_{\text{sp}}^k - T_o^k)}{\alpha_{3,c}^k} + \alpha_{3,d}^k \quad (17)$$

$$\alpha_{3,a}^k = \frac{(t_{T_z^{k-1}=T_{z,0}^{k-1} \pm 1}^{k-1} - t_{T_z^{k-1}}^{k-1})}{(T_{z,1}^{k-1} - T_{z,0}^{k-1})} \quad (18)$$

$$\alpha_{3,b}^k = \frac{(t_{T_z^{k-1}=T_{z,0}^{k-1} \pm 1}^{k-1} - t_{T_z^{k-1}}^{k-1})}{(T_z^{k-1} - T_{z,0}^{k-1})} \text{ when OA damper is closed and } \begin{cases} T_o^{k-1} > T_{\text{csp}}^{k-1} + \alpha_{3,c}^k \\ T_o^{k-1} < T_{\text{hsp}}^{k-1} - \alpha_{3,c}^k \end{cases} \quad (19)$$

$$\alpha_{3,d}^k = \begin{cases} \alpha_{3,d}^{k-1} + \alpha_{3,e}^k, & T_{z,\text{occ}}^{k-1} < T_{\text{sp}}^{k-1} - 1 \\ \alpha_{3,d}^{k-1} - \alpha_{3,e}^k, & T_{z,\text{occ}}^{k-1} > T_{\text{sp}}^{k-1} + 1 \end{cases} \quad (20)$$

In the first term of Eq. (17), the first parameter, $\alpha_{3,a}^k$ (minutes/1 °F), denotes the time required to change the indoor temperature from $T_{z,0}^{k-1}$ by 1 degree. In the second term, two parameters are used to consider the influence of the outdoor temperature changes on the indoor temperature. The second parameter, $\alpha_{3,b}^k$ (minutes/1 degree), represents the time interval required for the indoor temperature to change by 1 degree. The third term is a parameter, $\alpha_{3,d}^k$, to adjust the optimal start time, in minutes, when the difference between the indoor temperature ($T_{z,\text{occ}}^{k-1}$) and the set point temperature (T_{sp}^{k-1}) is ± 1 degree on the previous day at occupancy. This parameter changes by a specific amount ($\alpha_{3,e}^k$) based on its value in the previous day, $\alpha_{3,d}^{k-1}$, as shown in Eq. (20). If it exceeds a certain threshold, it resets to zero.

2.1.4. Model 4

The fourth model uses a simple first-order response model to predict response time [24]. The first-order overall response model has a simple closed-form solution that can be used to predict the time to reach a new set point, as shown in Eqs. (21) through (23). This model observes the

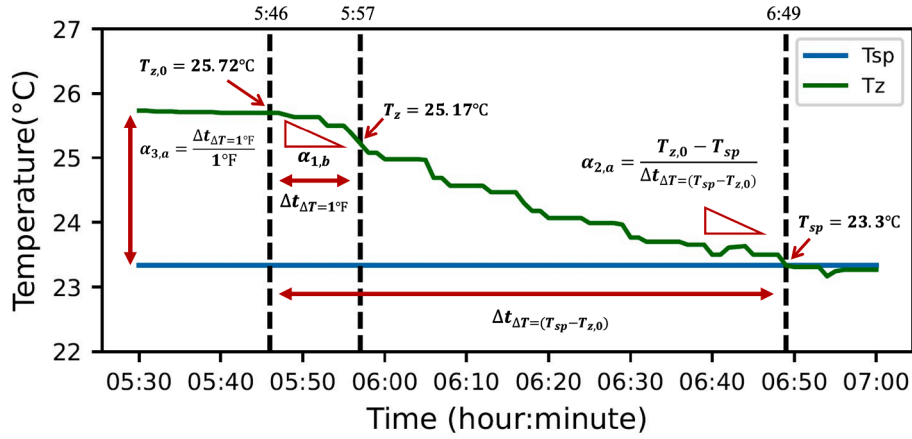


Fig. 2. Temperature slope calculation to estimate how long it takes for indoor temperature changes.

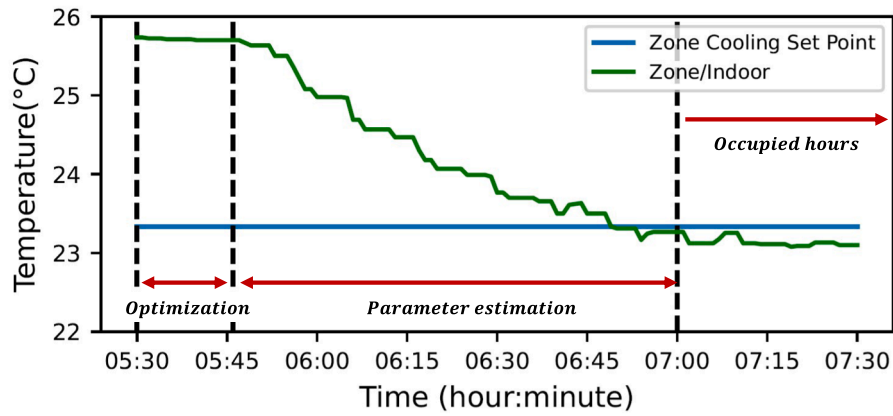


Fig. 3. Schedule of optimal start strategies.

response of the zone temperature after the RTU starts and continuously updates its parameters based on the deviation between the set point temperature and zone temperature. Eq. (21) formulates the optimal start time ($t_{4,opt}^k$) using three parameters. The first parameter ($\alpha_{4,a}$) is a constant value representing an acceptable dead band from the set point temperature, while the second parameter ($\alpha_{4,b}^k$) denotes the initial deviation between the set point and the zone temperature after the RTU starts as shown in Eq. (22). The last parameter ($\alpha_{4,c}^k$) reflects the changing conditions or system dynamics calculated based on the error variations over time after RTU starts. It uses the least squares approach

to gather historical error information and continuously adjust $\alpha_{4,c}^k$ until the error trend discontinues or becomes small enough. A large $\alpha_{4,b}^k$ or small $\alpha_{4,c}^k$ results in a longer start time, whereas a small $\alpha_{4,b}^k$ or large $\alpha_{4,c}^k$ leads to a shorter start time:

$$t_{4,opt}^k = \frac{\ln\left(\frac{\alpha_{4,a}}{\alpha_{4,b}^k}\right)}{\ln(\alpha_{4,c}^k)} \quad (21)$$

$$\alpha_{4,b}^k = (T_{sp}^{k-1} - T_{z,0}^{k-1}) \quad (22)$$

$$\alpha_{4,c}^k = \frac{\sum_{i=2}^N (T_{sp}^{k-1} - T_{z,i-1}^{k-1})(T_{sp}^{k-1} - T_{z,i}^{k-1})}{\sum_{i=2}^N (T_{sp}^{k-1} - T_{z,i-1}^{k-1})^2} \quad (23)$$

where N is a batch of samples to estimate $\alpha_{4,c}^k$.

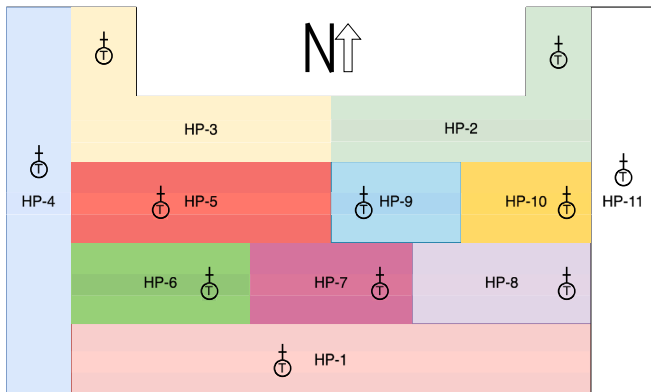


Fig. 4. Location of heat pump and the thermostats in Building A.

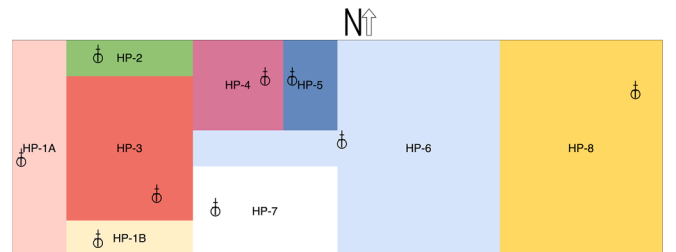
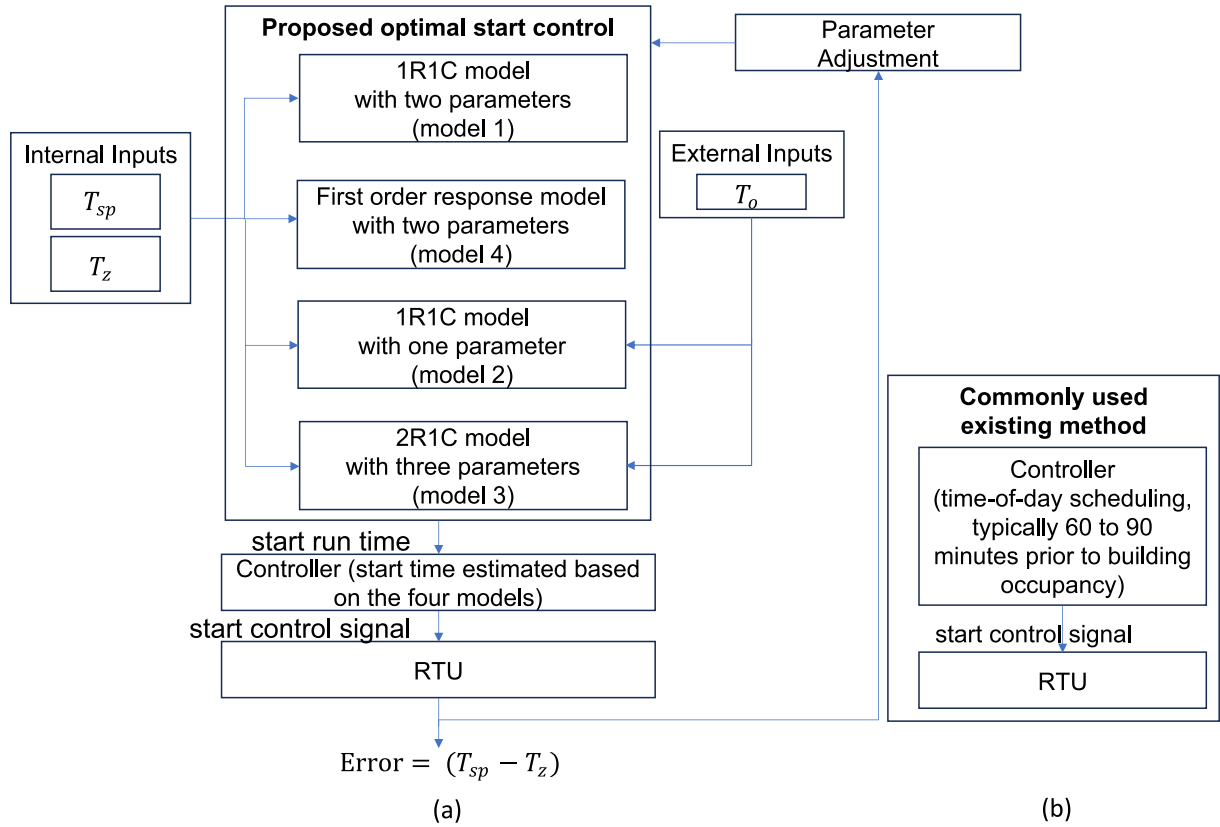


Fig. 5. Location of heat pump and thermostats in the Building B.

Table 1

Heat pump system power for Buildings A and B.

Building A				Building B			
Units	System Power [kW]	Number of Compressor	Room type	Units	System Power [kW]	Number of Compressors	Room type
HP1	6.6	1	Office	HP1A	2.0	1	Manager Office
HP2	6.7	1	Office	HP1B	2.0	1	Office
HP3	6.7	1	Office	HP2	2.0	1	Manager Office
HP4	6.7	1	Office	HP3	7.5	2	Lobby
HP5	6.6	1	Office & Meeting	HP4	7.5	2	Shop
HP6	7.7	1	Office	HP5	4.0	1	Shop
HP7	5.1	1	Meeting	HP6	25.7	2	Shop & Office
HP8	5.1	1	Kitchen	HP7	7.5	2	Office
HP9	5.1	1	Office	HP8	20.1	2	Shop
HP10	5.1	1	Office				
HP11	5.1	1	Office				

**Fig. 6.** Comparison of proposed optimal start control approach (a) with the commonly used existing start approach (b).

2.2. Parameter estimation

This section describes parameter identification for the optimal start algorithms based on historical data for the RTUs. The indoor temperature gradient over time, used across all models described in the previous sections, is referred to as the “temperature slope.”

Fig. 2 shows the indoor temperature changes over time when the zone is cooled before occupancy. In this case, the RTU needs 63 min to lower $T_{z,0}$ by 2.42 °C (4.3°F) to reach set point 23.33 °C (74°F). The temperature slope (e.g., $\alpha_{2,a}$, $\alpha_{3,a}$) indicates the rate at which the indoor temperature responds to zone load conditions (i.e., outdoor temperature, initial indoor temperature). The temperature slope varies daily, reflecting the dynamic interaction between outdoor and indoor conditions and its correlation with zone set point temperature and seasonal variations. This temperature slope plays a crucial role in determining the optimal start time.

The frequency of parameter updates is another critical factor in

ensuring the effectiveness of the optimal start methods. For instance, parameters tuned in the previous month may lead to non-optimal performance, as weather and indoor thermal conditions can vary significantly within that time frame. On the other hand, the parameters solely tuned with data from the previous day may lead to ineffective results when noticeable day-to-day thermal changes occur, resulting in substantial parameter fluctuations.

Thus, an exponential moving average was applied to the daily update parameters, allowing the models to capture both recent observations and historical trends in thermal conditions and environmental factors. This approach aims to achieve more accurate and reliable performance in dynamic conditions. Note that the choice of parameter tuning strategy should be tailored to the specific characteristics of the user’s environment. The exponential moving average update equation is expressed as:

$$\bar{\alpha}^k = \bar{\alpha}^{k-1} + \omega(\alpha^k - \bar{\alpha}^{k-1}) \quad (24)$$

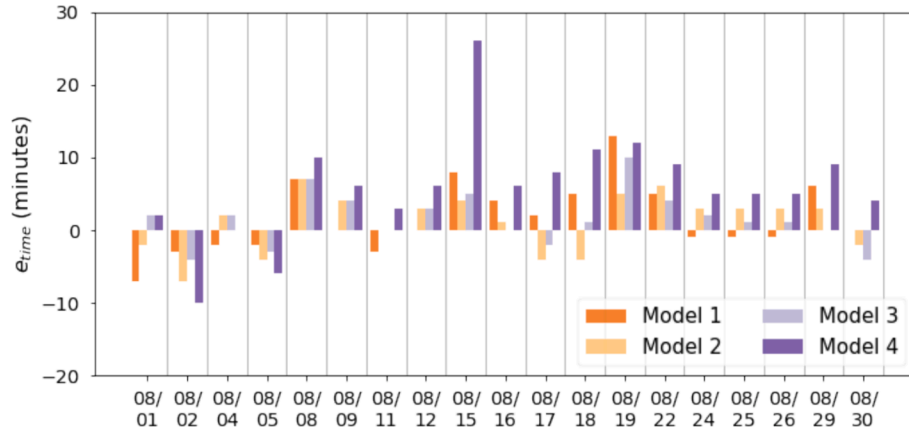


Fig. 7. Time-based accuracy error (e_{time}) for HP2 within summer season (August): Positive error means the estimated optimal start times are short, while the negative error means they are long for the zone to reach the set point temperature.

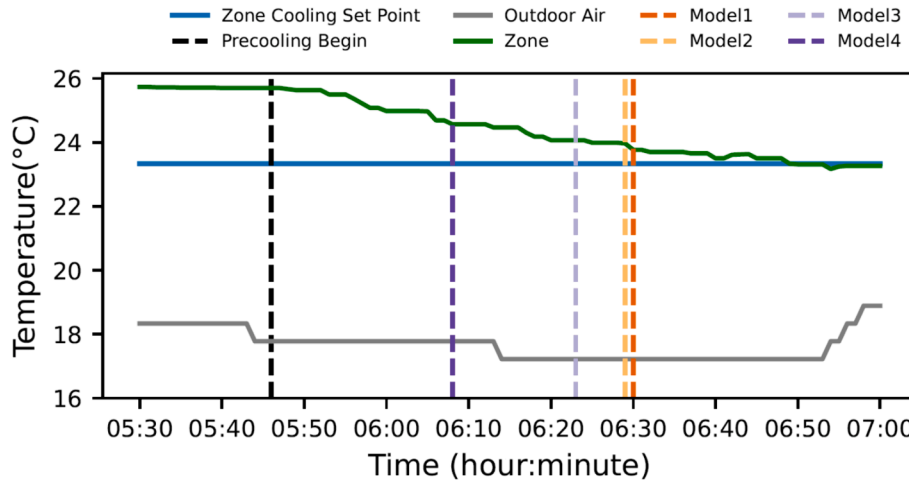


Fig. 8. One-day snapshots of HP2 (8/15) with zone temperature, cooling setpoint, outdoor air temperature, and the estimated precooling end times for the four models.

where ω is a forgetting constant, determining the degree to which new observations influence the updated parameter values. The model parameters are adjusted through training using historical data. Typically, the training dataset includes the last 10 weekdays of rolling data.

2.3. Cost function

The pre-conditioning optimization problem is an integer linear program concerned with finding the RTU on/off control signal over a morning period that minimizes energy cost subject to maintaining the indoor air temperature within bounds. Typically, RTU control signal u^k is determined by a thermostat that attempts to control T_z^k around a set point T_{sp}^k . To avoid high-frequency cycling, a dead band of width $\pm\sigma$ is often inserted, leading to a hysteresis behavior. For the cooling mode, this hysteresis behavior can be expressed as:

$$u^k = \begin{cases} 1 & \text{if } u^{k-\delta} = 0 \text{ \& } T_z^{k-\delta} > T_{sp} + \sigma \\ 0 & \text{if } u^{k-\delta} = 1 \text{ \& } T_z^{k-\delta} < T_{sp} - \sigma \\ u^{k-\delta} & \text{otherwise} \end{cases} \quad (25)$$

where $k-\delta$ is the ‘previous’ time instant based on which u^k is determined.

Determination of the system start time can be expressed as an optimization problem. The continuous-time problem can be reformulated in

discrete-time as follows: find u^k for $k = 1, 2, \dots, m$ that minimizes the objective function.

$$\min_{u^k \in \{0,1\}^m} \sum_{k=t_0}^{t_0+\Delta t} u^k[k] \quad (26)$$

Subject to the constraints:

$$-\sigma \leq (T_z^k - T_{sp}^k) \leq \sigma \quad (27)$$

$$t_{\min} \leq t_z^k \leq t_{\max} \quad (28)$$

where the objective is to minimize the runtime of the system subject to constraints. This method of determining the optimal RTU on/off control sequence $u^*[j]$ for $j = 1, 2, \dots, m$ is referred to as the proposed optimal start strategy.

After $u^*[j]$ is calculated, the corresponding optimal start time $t_z^*[j]$ can be calculated. With $u^*[j]$ and the models described in Section 2.1 Models to Predict Response Time, the corresponding optimal start time $t_z^*[k]$ for $k = 1, 2, \dots, m$ can be calculated as is done in Section 4. **Results and Discussion** The first constraint ensures that the set point error at the desired occupancy time is equal to or less than the target temperature with a dead band (σ). The second constraint ensures that $t_{\max}$ and $t_{\min}$ are defined as the maximum and minimum times allowed for the equipment to run before occupancy. This value is included so the user

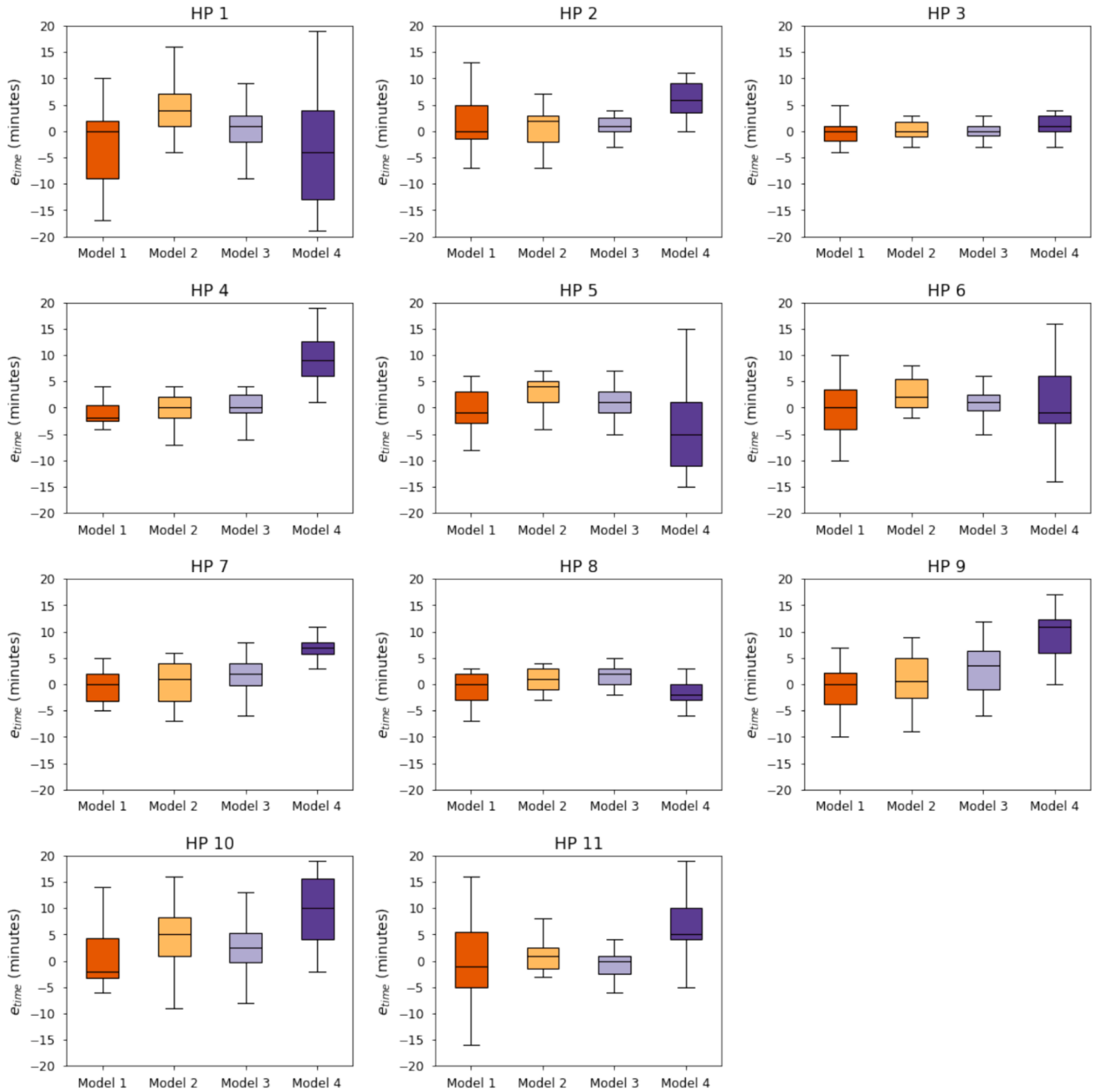


Fig. 9. Boxplot of time-based accuracy error (e_{time}) for the four optimal start algorithms: Positive error means the estimated optimal start times are short, while negative error means they are long for the zone to reach the set point temperature.

can put (safety) constraints around the algorithm to prevent it from starting the systems too early. The time to reach the occupied set point is calculated.

2.4. Summary of methodology

The optimal start strategies use learning algorithms and real-time measurements to minimize RTU runtime while the indoor temperature reaches the set point temperature.

Each day, at a specific time (i.e., the earliest start time as configured by building operation staff), the optimal start algorithm begins by collecting the outdoor and zone temperatures. Using these temperatures, it calculates the optimal start interval, as illustrated in Fig. 3. This interval is subtracted from the scheduled start time to determine the target time. The control system then monitors the zone temperature to adjust the actual start interval needed to reach the desired heating or cooling set point.

Once the RTU is started during the optimal start period, the parameter estimation stage begins by recording the time duration for the indoor temperature to reach the set point temperature. When the timer is initiated, the zone temperature is sampled, and the temperature slope is approximated using method from each model.

After computing the temperature slope for each day, the moving average of the temperature slope during the previous days is calculated. At a certain time before the occupancy each day, another sample of the zone temperature is taken, and the optimal start time is calculated using each model. The learning process is carried out at night based on the historical data of recovery operation before optimal start operation. The moving average of the temperature slope from past days is used to predict the optimal start time for the current day. Separate parameters are maintained for zone temperature outdoor air temperature, and optimized start time for both heating and cooling.

It is important to note that some (e.g., one week) training data is required before an accurate optimal start time can be determined. This

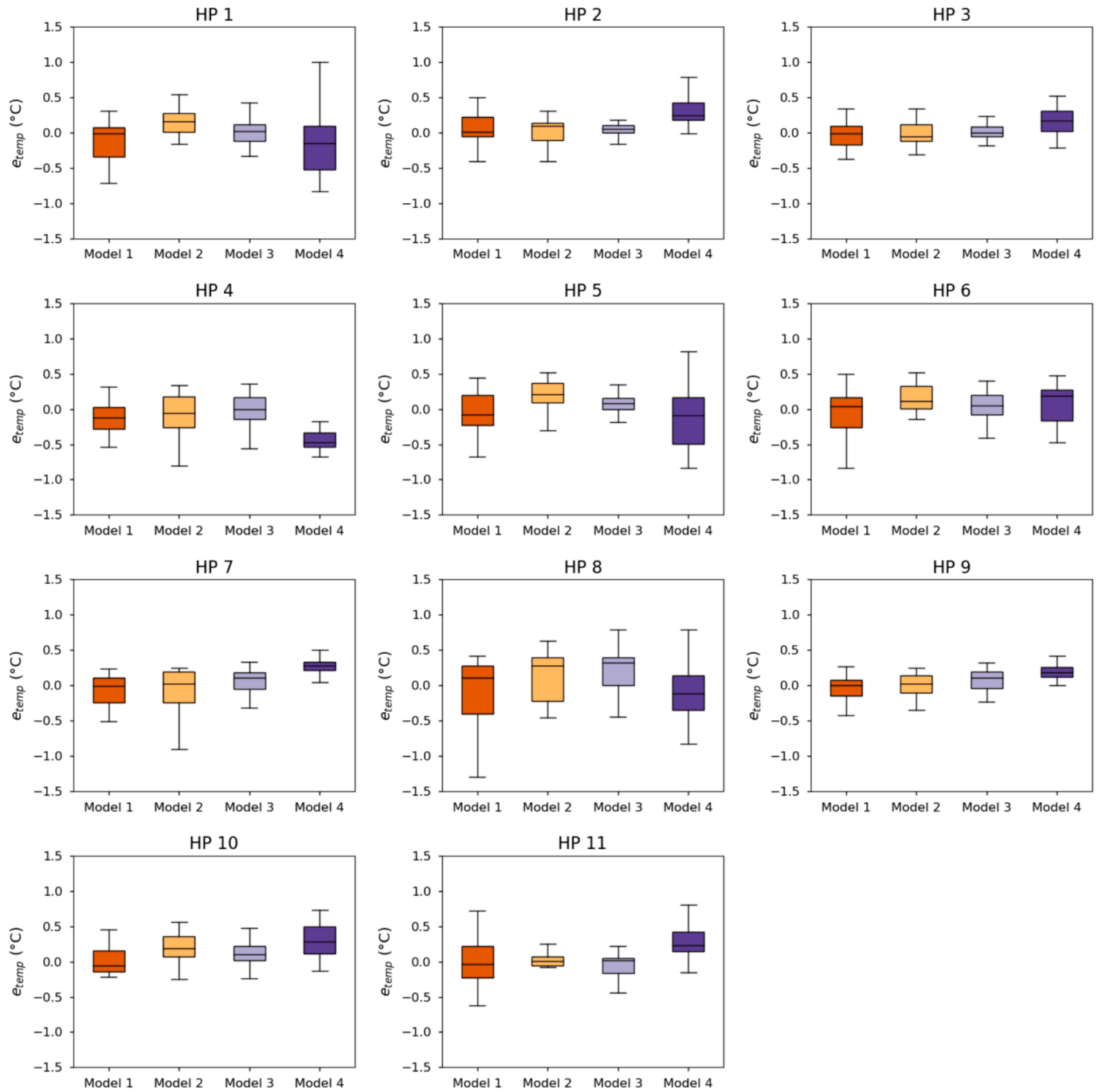


Fig. 10. Boxplot of temperature-based accuracy error (e_{temp}) for the four optimal start algorithms: Positive error means the estimated temperatures with optimal start are higher, while negative error means they are lower than the temperature set point.

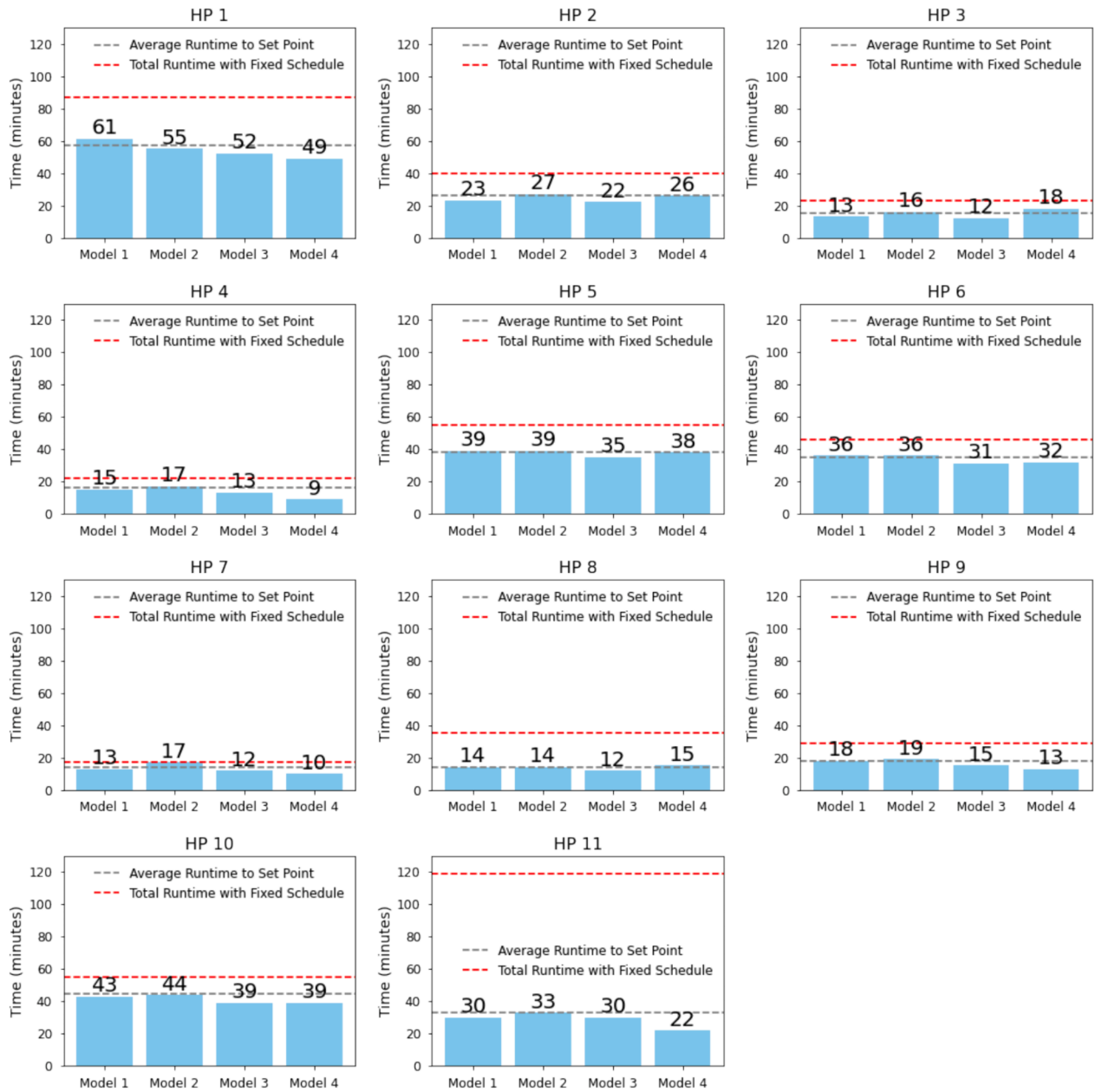


Fig. 11. Average optimal start control runtime of all units in Building A during summer (July-August).

Table 2

Average energy savings per day with the optimal pre-cooling in Building A.

	Average Energy Savings (kWh/day)				Average Energy Savings (%)			
	Model 1	Model 2	Model 3	Model 4	Model 1	Model 2	Model 3	Model 4
HP1	2.9	3.6	3.9	4.3	30	37	40	45
HP2	2.0	1.5	2.1	1.6	44	33	46	35
HP3	1.0	0.7	1.2	0.5	42	30	46	20
HP4	0.8	0.5	1.0	1.5	32	22	41	60
HP5	1.7	1.7	2.2	1.8	28	29	37	31
HP6	1.3	1.3	2.0	1.9	21	23	34	31
HP7	0.3	0	0.5	0.6	24	1	32	40
HP8	1.8	1.8	2.0	1.7	62	59	67	58
HP9	0.9	0.9	1.2	1.3	36	36	50	55
HP10	1.0	0.9	1.3	1.3	21	20	28	28
HP11	7.6	7.3	7.6	8.2	75	72	75	81

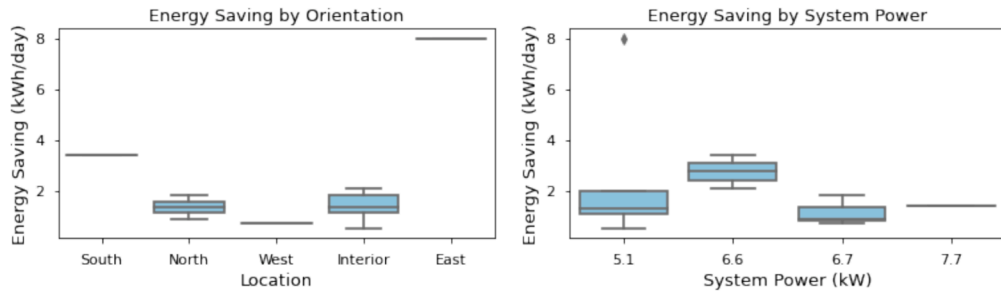


Fig. 12. Energy savings with optimal start control by orientation and system power in Building A during summer.

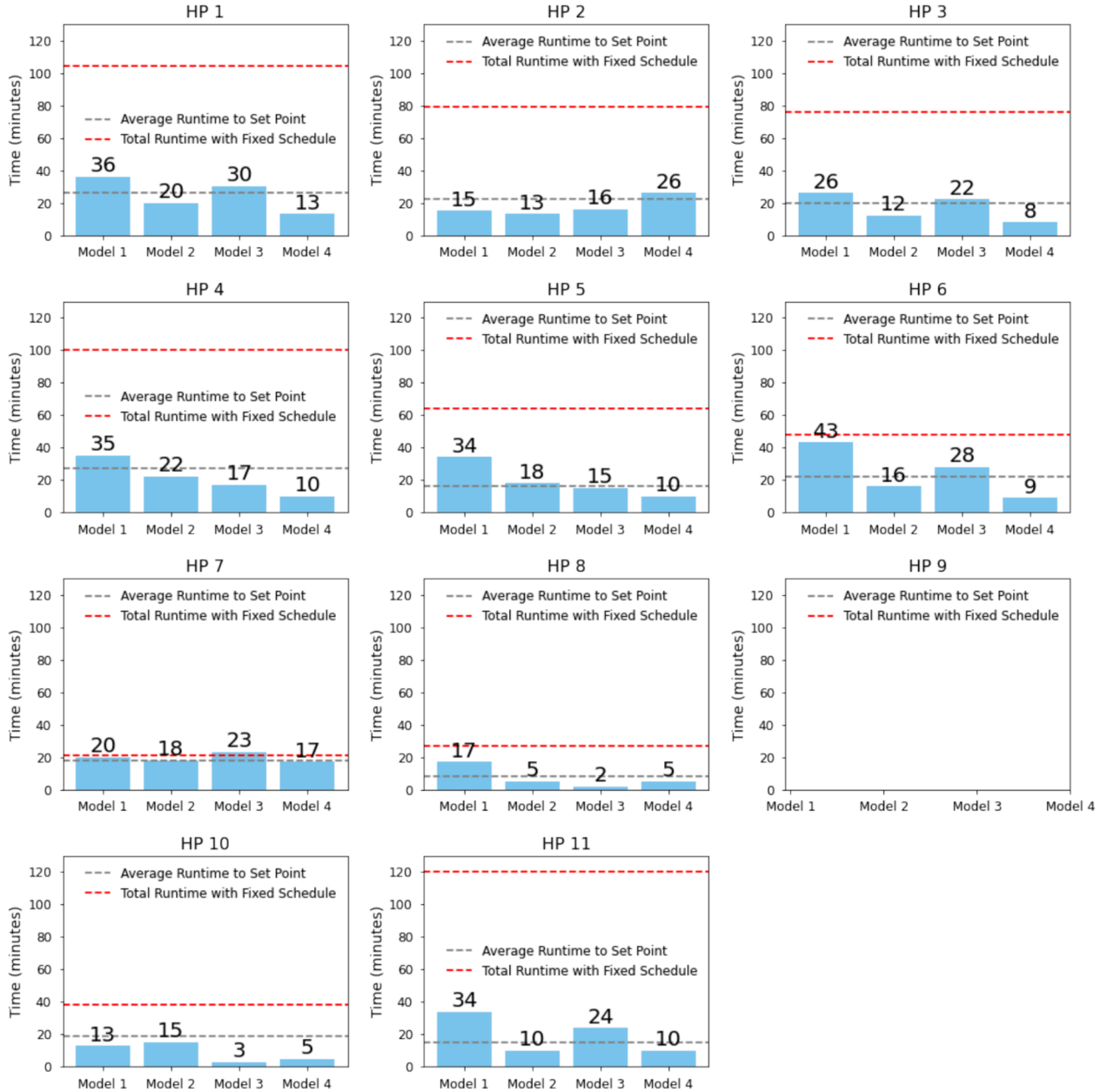


Fig. 13. Average optimal start control runtime of all units in Building A during winter (November–February).

training is necessary to adjust the parameters based on the historical data. When optimal start is initially deployed, the optimal start time defaults to the maximum configured value for precooling/preheating.

2.5. Accuracy evaluation

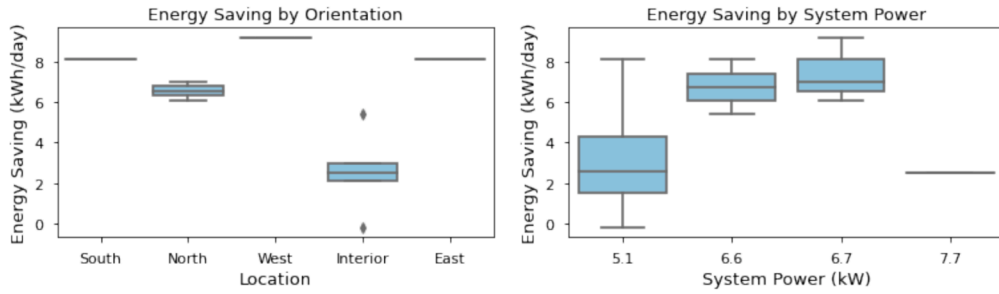
2.5.1. Accuracy evaluation with optimal start time

The first approach to assessing the accuracy of the optimal start

Table 3

Average energy savings per day with the optimal pre-heating in Building A.

	Average Energy Savings (kWh/day)				Average Energy Savings (%)			
	Model 1	Model 2	Model3	Model 4	Model 1	Model 2	Model 3	Model 4
HP1	7.4	9.2	8.1	10.0	65	81	71	88
HP2	7.1	7.4	7.0	6.0	81	83	79	68
HP3	5.6	7.2	6.1	7.6	66	85	72	89
HP4	7.3	8.7	9.2	10	65	78	83	90
HP5	3.2	5.1	5.4	5.9	46	72	77	84
HP6	0.6	4.1	2.5	5.0	10	66	41	81
HP7	0.1	0.2	−0.2	0.3	6	14	−8	19
HP8	0.9	1.9	2.1	1.9	39	83	92	81
HP9	0	0	0	0	0	0	0	0
HP10	2.1	2.0	3.0	2.8	66	61	92	87
HP11	7.3	9.3	8.1	9.4	72	91	80	92

**Fig. 14.** Energy savings with optimal start control by orientation and system power in Building A during winter.

control is to evaluate the time error (e_{time}) between the estimated pre-operation time (t_{opt}^k) obtained from the optimal start methods and the actual operation time (t_{act}^k) required for the RTU to warm up or cool down the indoor temperature to set point. This metric shows how closely the optimal start solutions align with the actual operation in terms of time.

$$e_{\text{time}} = t_{\text{act}}^k - t_{\text{opt}}^k \quad (29)$$

When the time error is a positive value, it indicates that the optimal start time is shorter than the operation time required to warm up or cool down the indoor temperature condition to the set point. In this case, the indoor condition might not be satisfactory to occupants. On the other hand, when the time error is a negative value, it indicates that the optimal start control time is longer than the required operation time. In this case, energy consumption would increase due to the extended operation as the indoor temperature would reach the temperature set point before the start of occupancy.

2.5.2. Accuracy evaluation with expected temperature

Another approach is to estimate the expected indoor temperature when operating for the time determined by the optimal start control, and

then to compare it with the set point temperature. Since it is not possible to know the actual temperature, the expected temperature by the optimal start solution is estimated based on the rate of 1 degree change per unit time ($\alpha_{3,a}^k$). Using this value, the expected temperature can be estimated assuming the zone is being cooled or heated during the optimal precool/preheat time from the initial indoor temperature. The following equations represent the formulation to calculate the expected temperature and temperature error.

$$T_{\text{opt-est}}^k = T_{z,0}^k \pm \alpha_{3,a}^k \times t_{\text{opt}}^k \quad (30)$$

$$e_{\text{temp}} = T_{\text{opt-est}}^k - T_{\text{sp}}^k \quad (31)$$

Note that actual operation may stop heating/cooling when the indoor temperature reaches the set point temperature; however, this expected temperature calculation does account for this thus the expected temperature may go beyond the set point temperature.

2.6. Energy savings calculation

To demonstrate the effectiveness of the optimal start control, the energy savings are estimated compared to the original pre-operation

Table 4

Average energy savings per day with optimal pre-cooling in Building B.

	Average Energy Savings (kWh/day)				Average Energy Savings (%)			
	Model 1	Model 2	Model 3	Model 4	Model 1	Model 2	Model 3	Model 4
HP1A	0.4	0.4	0.4	0.3	46	43	43	35
HP1B	0.1	0.1	0.1	0.5	11	11	6	48
HP2	0.3	0.5	0.3	0.5	22	31	20	36
HP3	3.2	2.4	2.8	2.7	34	26	30	28
HP4	2.3	2.0	2.3	1.4	53	45	51	32
HP5	2.0	1.9	2.2	3.3	43	41	49	72
HP6	12.3	10.2	12.5	13.2	67	55	68	72
HP7	2.0	2.0	3.2	2.2	65	66	99	70
HP8	21.8	18.7	28.2	31.0	54	47	70	78

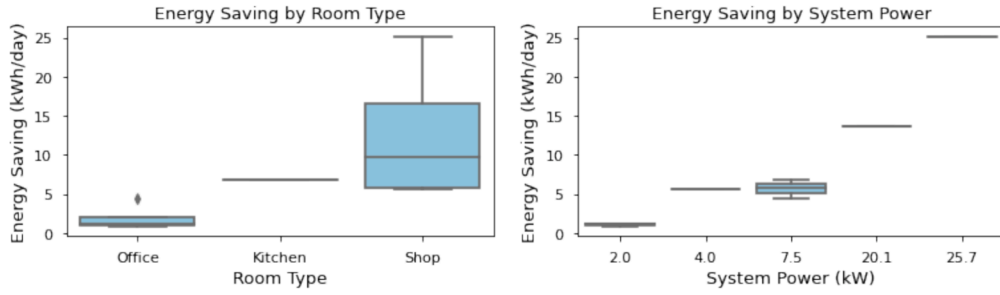


Fig. 15. Energy savings with optimal start control by room type and system power.

time (fixed schedule). The potential energy savings ($\Delta \text{kWh}_{\text{gross}}$) by using optimal start control are calculated by first computing the difference in runtime of the HP unit based on the baseline (fixed schedule) and the runtime estimate based on the optimal start. Then, the kilowatt-hours savings are computed by multiplying the rated capacity of the unit and the runtime difference. This comparison between the estimated optimal operating time and the measured operating time, as detailed in Eq. (32), highlights the energy saving potential of the optimal start control.

$$\Delta \text{kWh}_{\text{gross}} [\text{kWh}] = \sum_{i=1}^n (\text{SysPower}_i [\text{kW}] \bullet (\text{Time}_{\text{base},i} [\text{h}] - \text{Time}_{\text{opt},i} [\text{h}])) \quad (32)$$

where SysPower_i is a system power of the unit, $\text{Time}_{\text{base}}$ and Time_{opt} are the operating hours without and with optimal start control, respectively.

3. System specifications and test conditions

The field data from 20 heat pumps (HPs) from two SMBs were used to evaluate and validate the optimal start algorithms. After successful completion of the offline evaluation, the experiment was designed and deployed to demonstrate how the optimal start algorithms can be implemented to minimize RTU runtime and to mitigate peak electricity energy consumption. The demonstration results showed that the optimal start algorithm can reduce energy consumption without impacting

occupant comfort.

3.1. System specification

Fig. 4 and Fig. 5 show how the HPs serve the various zones in the two buildings. In Building A, most HPs serve the office, meeting rooms, and kitchen areas. In Building B, the HPs serve the office, the shop, and kitchen areas. As shown in Fig. 5, the HPs with smaller capacities serve the office area while those with larger capacities serve the shop.

The details of the HPs serving the two SMBs are given in Table 1. In Building A, all HPs have only one refrigerant circuit with one compressor and one condenser fan with an electric backup heater. In Building B, four HPs have one refrigerant circuit with one compressor with backup electric heater. Five HPs have two refrigerant circuits with two compressors and two condenser fans with backup electric heater.

3.2. Test conditions

The two test buildings use a BAS to control the HPs. Because the optimal start algorithms could not be directly deployed in the BAS, an open-source distributed sensing and control Internet-of-things platform VOLTTRON™ is used [25]. VOLTTRON provides access to key variables such as zone indoor temperature, zone temperature set point, and the compressor commands for the HPs; it also provides a means of indirectly controlling the HPs by changing zone set points. The optimal start algorithms are deployed in the two buildings using the VOLTTRON

Table 5

Average energy savings per day with optimal start control in Building B during winter.

	Average Energy Savings (kWh/day)				Average Energy Savings (%)			
	Model 1	Model 2	Model 3	Model 4	Model 1	Model 2	Model 3	Model 4
HP1A	0.8	1.0	0.7	0.7	25	30	19	21
HP1B	0.4	0.6	0.4	0.6	11	17	10	16
HP2	0.7	1.0	0.9	0.6	20	26	24	17
HP3	1.5	1.8	1.6	2.4	16	19	17	25
HP4	2.2	3.6	1.7	3.3	23	39	18	35
HP5	2.5	2.4	2.7	2.4	45	43	49	43
HP6	4.6	6.7	7.8	7.3	39	56	66	62
HP7	1.4	2.3	3.6	3.1	35	58	94	77
HP8	—	—	—	—	—	—	—	—

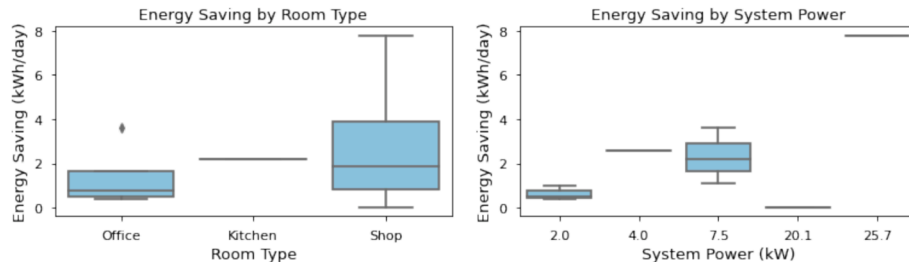


Fig. 16. Energy savings with optimal start control by zone type and system power.

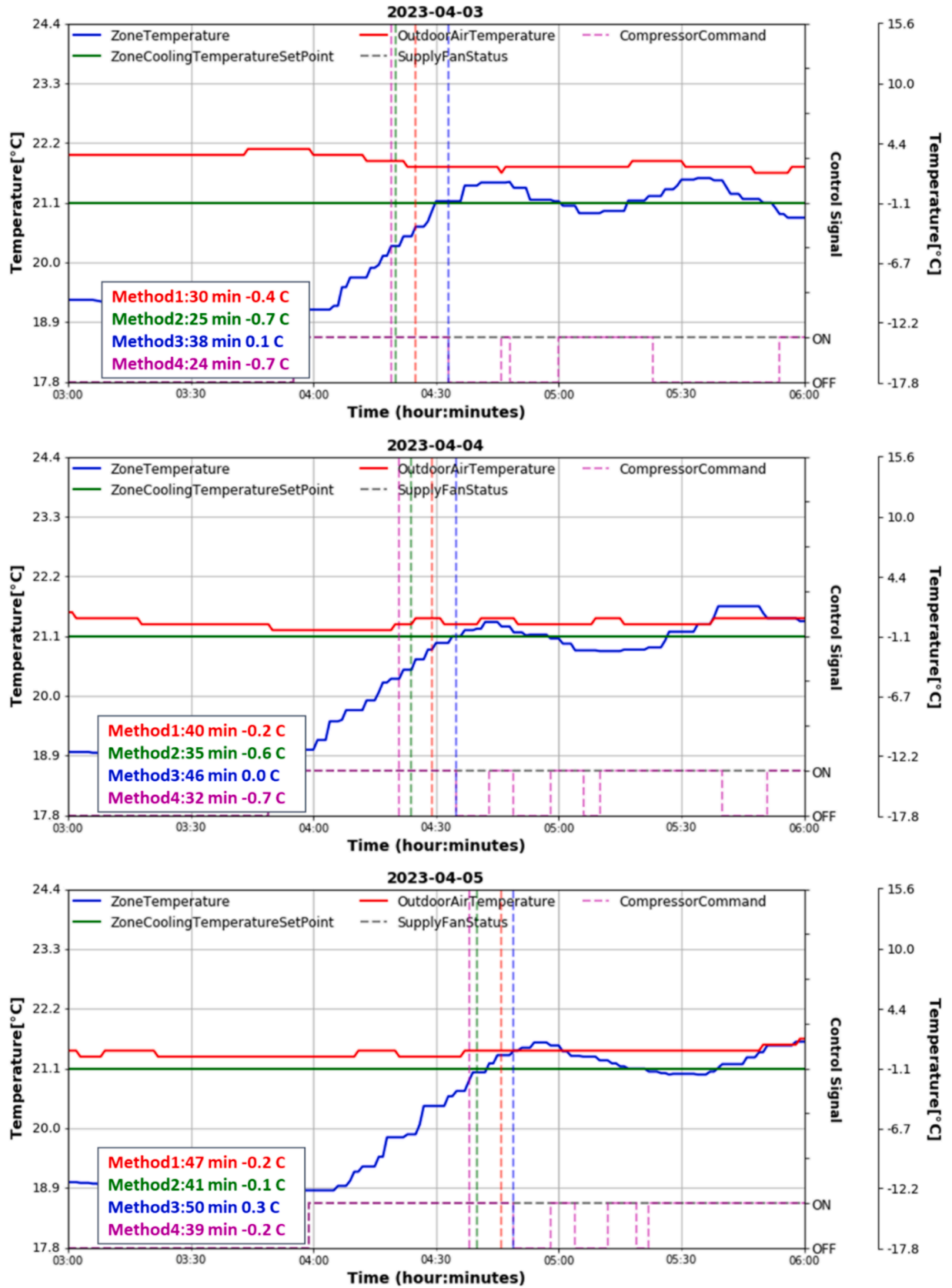


Fig. 17. Time series plots for a selection of an RTU showing how the optimal start algorithms activate the zone at different times during heating season.

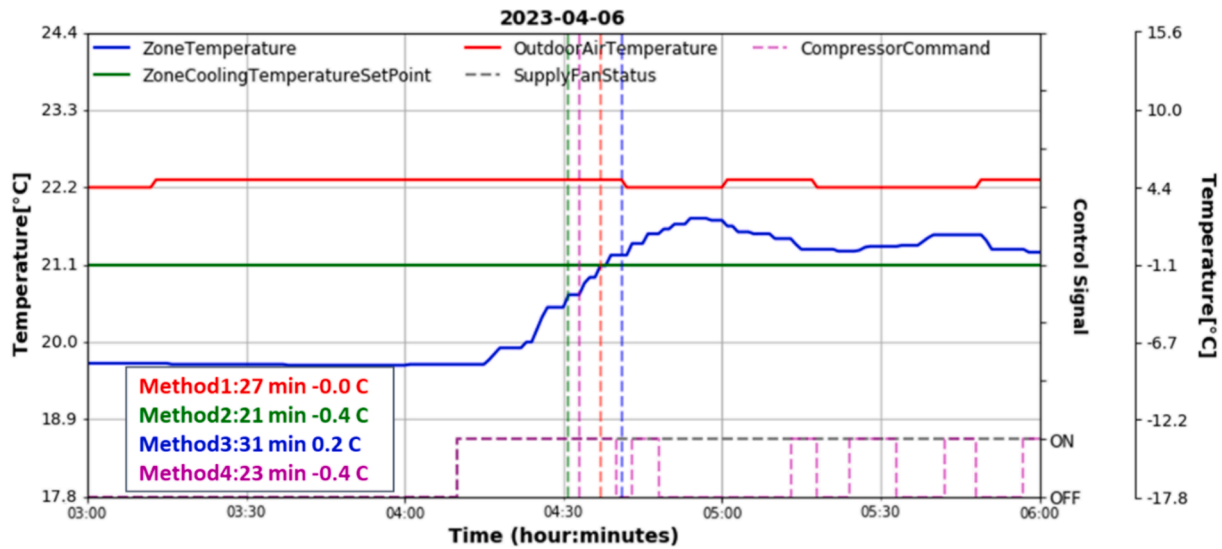


Fig. 17. (continued).

platform. Some integration development and testing were performed to obtain the desired communications functionality between VOLTTRON running the optimal start algorithms (coded in Python) and the BAS. After successful completion of the initial tests, additional functional testing of the optimal start algorithms was performed to verify safe operation. This time is then compared with the time remaining before occupancy and the set point is changed to the occupied value when the predicted time is less than or equal to the time remaining.

The four optimal start methods use two sensor measurements: zone temperature and outdoor air temperature. The accuracy of these temperature sensors is 0.56 °C. The frequency of data collection is 1-minute.

The performance of the optimal start control methods was evaluated during the summer (i.e., 7/1 ~ 8/31) and winter (i.e., 11/1 ~ 2/28) seasons considering two factors: accuracy and energy savings. The analysis was based on the weekday data alone; weekend and holiday data were excluded. The evaluation datasets included zone temperature and set point, supply fan status, cooling/heating commands, and outdoor air temperature. Fig. 6 compares the proposed optimal start method approach with the commonly used existing control approach.

4. HYPERLINK “SPS:id::Sec4” Results and Discussion

This section describes the performance assessment of the optimal start control algorithms using both existing field test data and real-time experimental data. The existing field data is historical archived data that is used to validate the optimal start methods in an offline environment. The real-time experimental data is data that is used in real-time from the various HPs to validate the optimal start methods in an actual deployment environment. The evaluation includes estimates of energy savings as well as accuracy of the algorithms.

4.1. Evaluation of optimal start control performance from existing field test data

The energy savings achieved by HP units can vary based on several factors, such as the location of the zones served by the HP (whether they are interior or exterior, oriented on the east or the west), the unit's capacity, and the zone type.

4.1.1. Accuracy evaluation

Fig. 7 shows the sample daily time-based accuracy error of the four optimal control methods for HP2 in Building A during the summer season (August). This information is used to investigate the magnitude

and direction of error for each optimal start method over time. Using the historical data, the optimal start time was estimated using all four optimal start algorithms. In Fig. 7, the positive error indicates that the estimated optimal start times are shorter than the time required for the zone temperature to reach the zone set point, whereas the negative error indicates that the estimated optimal start times exceed the time required for the zone temperature to reach the set point.

All four methods performed well in estimating the optimal start time, with exception of 8/15, where Model 4 had a time error exceeding 20 min (under-predicted the start time). To further investigate the relatively large accuracy error, a 1-day snapshot of HP2 operation on 8/15 (Fig. 8), which is a Monday, were analyzed. Fig. 8 shows the zone temperature, zone temperature set point, outdoor air temperature, and the compressor command signals. Because the unoccupied zone temperature (25.56 °C / 78 °F) is significantly higher than the occupied zone set point (23.33 °C / 74 °F) and the previous 2 days were unoccupied, the thermal mass of the zone is close to the unoccupied zone set point. The HP needs more time to cool down the thermal mass due to the building's thermal inertia which is influenced by the building construction. As the building has warmed up over the previous two days, this stored heat must be removed before reaching the desired setpoint, which generally takes longer. Despite the lower outdoor air temperatures during the night and early morning in the location, the thermal mass continues to radiate the heat inward, complicating the cooling process during the subsequent day. Consequently, when cooling operations resume with the occupied setpoint temperature, the HVAC system requires more time to lower the zone temperature to set point. Most models, especially Model 4, did not account for operation, resulting in a longer time needed to cool the thermal mass to the zone temperature set point. Therefore, the methods sometimes may suggest a shorter optimal start time, which could result in relatively larger errors as observed on most Mondays. This result highlights the need to consider a different model for the days after weekends and holidays.

Fig. 9 and Fig. 10 are boxplots of the time-based and temperature-based accuracy errors for the four optimal start control algorithms, providing the distribution and magnitude of the errors. Even though there are a few time errors that exceed 15 min, all methods showed a mean time-based error of less than 10 min, indicating good reliability and consistency in their performance. Temperature error is defined as the difference between the zone temperature and occupied zone temperature set point at the occupancy start time. All methods showed a mean temperature-based accuracy error below 0.5 °C, indicating acceptable temperature error as well.

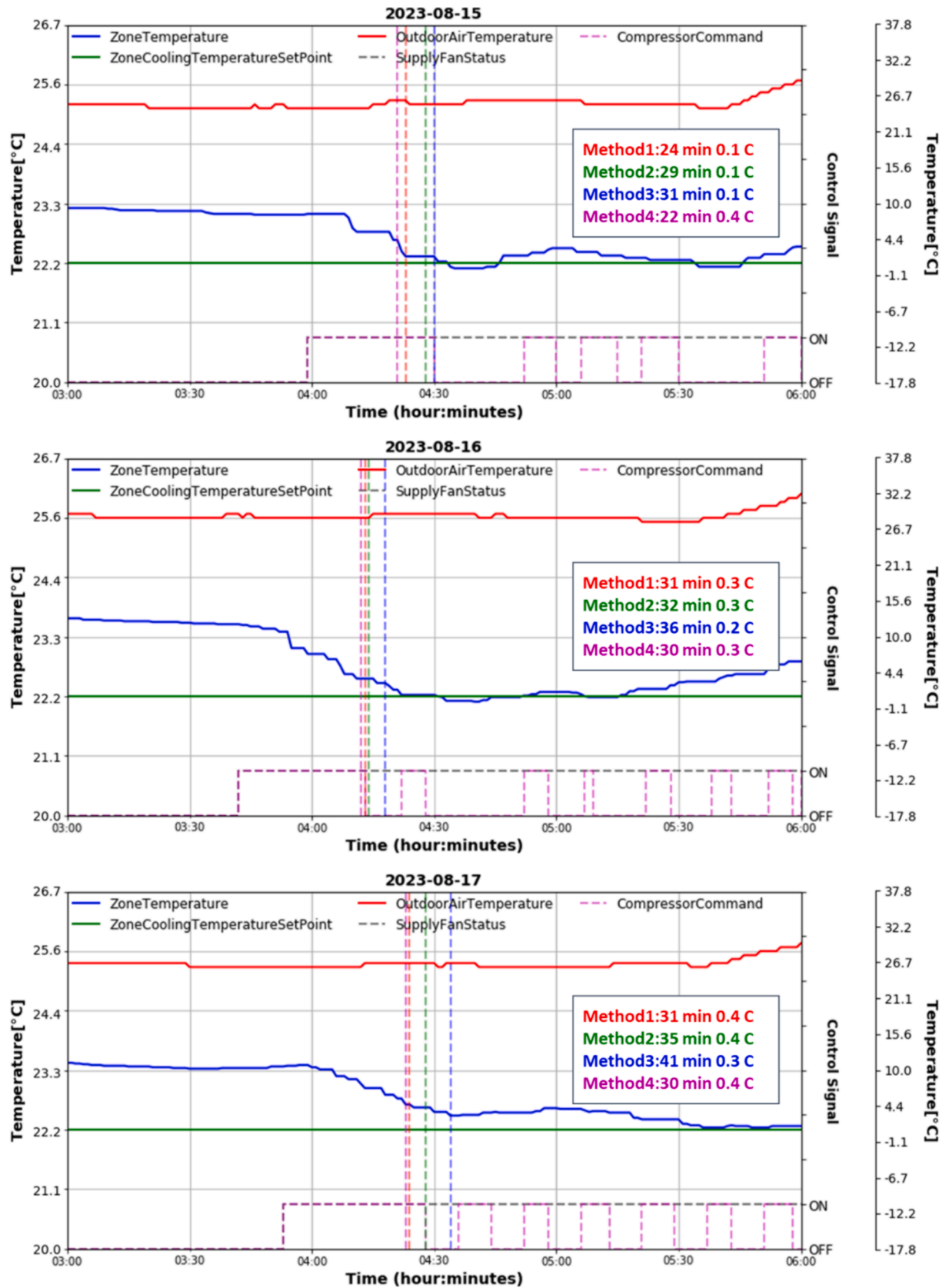


Fig. 18. Time series plots for a selection of an RTU showing how the optimal start algorithms activate the zone at different times during summer season.

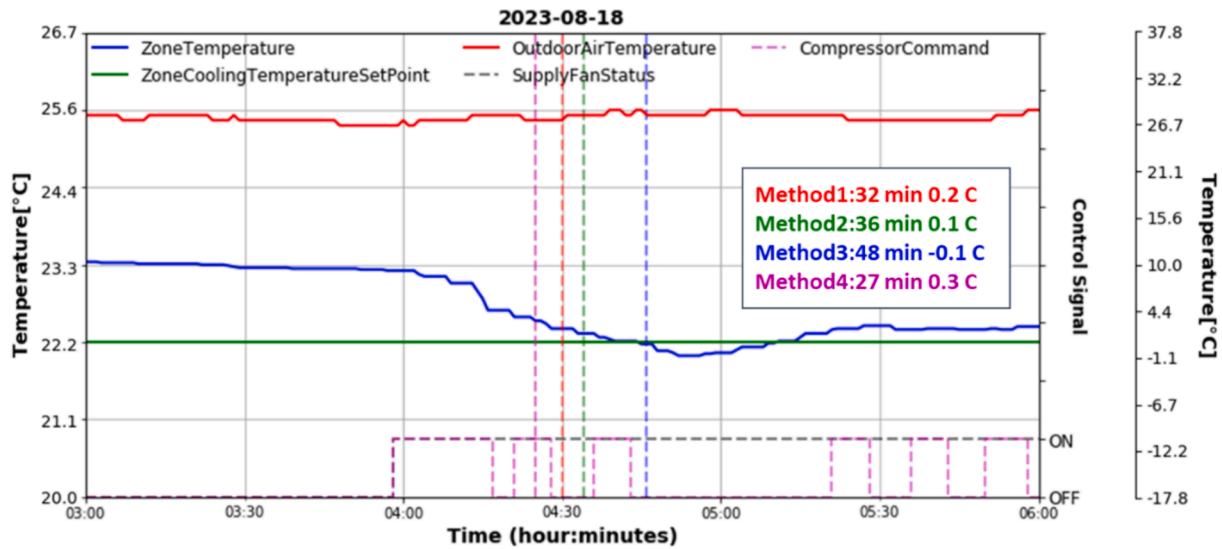


Fig. 18. (continued).

4.1.2. Energy savings estimates from optimal start in building a

The potential energy savings compared to fixed schedule-based operation can be estimated for the optimal start field test for Building A. This section describes the energy-saving potential from the use of optimal start controls for both pre-cooling and pre-heating for Building A.

Fig. 11 shows the average optimal start runtime (histograms) for each of the four models for all units in Building A during the summer season (July–August). The graph also shows the average time when the zone temperature reached the zone set point (gray dashed line) and the total runtime (red dashed line) before occupancy when the fixed runtime (baseline) was employed.

Based on the information presented in Fig. 11, the average runtime avoided with optimal start control can be estimated along with the corresponding average daily energy savings, expressed as a percentage of the total runtime with a fixed schedule as shown in Table 2. Each HP unit operates uniquely for its respective zones, resulting in different baseline energy consumptions and energy saving potential. HP11 has a fixed pre-operation schedule set 2 h before occupancy, which is longer than the other units, resulting in significant run-time time savings from optimal start control. Using optimal start controls, the average pre-operation time can be reduced by 22 min across all units compared to the existing fixed schedule strategy, potentially leading to an energy savings of 40 % on average across all units.

HP1, HP2, HP3, HP4, and HP11 serve exterior zones in Building A, while HP5, HP6, HP7, HP8, HP9, and HP10 serve interior zones. HP1 is situated on the south side, HP11 on the east side, and HP4 on the west side. HP2 and HP3 serve the north side of the building. Classifying the units based on their system power/capacity, HP7, HP8, HP9, HP10, and HP11 have the lowest capacity at 5.1 kW; HP1, HP2, HP3, HP4 and HP5 have a capacity ranging between 6.6 and 7 kW; and HP6 has the largest capacity at 7.7 kW.

To provide insights into how energy savings vary across different locations and power capacities, Fig. 12 shows grouped bar and boxplots of the average energy savings by the zone location that the HP serves and the HP system power. For instance, the HPs serving exterior zones are exposed to varying weather conditions, and these units (HP1, HP2, HP3, HP4, and HP11) may experience different temperature conditions based on their orientation (south, north, west, and east). As observed in Fig. 12, HP11, serving east side of Building A, shows significant energy savings, which is approximately 3.5 times larger than those of the interior units and around twice as large as any other units serving exterior zones.

Units serving the east side of the building may receive more direct sunlight, leading to increased cooling demands and significant benefits of optimal start control. Units serving interior zones (HP6, HP7, HP8, HP9, and HP10) show more variability and lower average energy savings compared to the units serving exterior zones. The energy-saving potential of units serving interior zones might be influenced by the temperature of adjacent rooms within the building. The capacity of the HP unit is also a significant factor; higher capacity units save more energy with the optimal start controls.

The energy savings from optimal start achieved by HPs also varied based on the season. Fig. 13 shows the average optimal start runtime (histogram) for each of the four models for all units in Building A during the winter season (November–February).

Table 3 summarizes the energy savings from optimal start control for pre-heating operation of all units in Building A. The energy savings from optimal start control for pre-heating are considerably greater when compared to pre-cooling. This is due to different conditions governing cooling and heating operations. Significant differences in outdoor temperature between heating and cooling operations might have contributed to the variations in energy-saving potential as well. During winter, the lower outdoor temperatures and higher difference between indoor and outdoor conditions result in increased energy consumption and optimal start savings compared to cooling. In Table 3, the negative value of Model 3 for HP 7 means that Model 3 did not save energy compared to the baseline. HP7 serves the interior of the building, which requires shorter preheating times compared to exterior units. As shown in Fig. 13, the average runtime for the baseline is 18 min, while the average runtime for Model 3 is 23 min, which is a difference of only 5 min. Therefore, the slight increase in average energy usage due to the additional runtime is not significant. For HP9, which also serves an interior zone, there was no need for additional preheating during the winter. The average zone temperature was already higher than the setpoint temperature because adjacent zones began heating earlier, leading to a heat transfer from these zones. Consequently, HP9 did not need to run for preheating, resulting in no measurable energy savings.

Fig. 14 shows the average energy savings with optimal pre-heating by the HP zone orientation and the HP system heating power during winter. Like cooling, the exterior units (HP1, HP2, HP3, HP4, HP5, and HP11) experience greater exposure to outdoor temperature variations, leading to increased energy savings compared to the units serving interior zones. Units serving interior zones (HP6, HP7, HP8, HP9, and HP10) show more variability and lower average energy savings compared to the units serving exterior zones. Additionally, the bigger

HP systems (HP1, HP2, HP3, HP4, HP5, and HP6) save more energy on average compared to the smaller HP systems (HP7, HP8, HP9, HP10, and HP11).

4.1.3. Energy savings analysis of optimal start methods for building B

Table 4 summarizes the energy savings from optimal start control for all units in Building B. Like Building A, energy savings between 40 % and 52 % on average can be achieved from optimal start by reducing the runtime. Fig. 15 shows the average daily energy savings categorized by zone type and system power.

HP1 and HP2 have the lowest capacity at 2 kW; HP5 is 4 kW; HP3, HP4, and HP7 have a capacity of 7.5 kW; HP8 operates at 20.1 kW; and HP6 is the largest with a capacity of 25.7 kW. Except for HP3, all HPs serve exterior zones. HP1, HP2, and HP7 serve office spaces; HP3 serves the kitchen; and HP4, HP6, and HP8 serve fabrication and carpentry shops. Examining energy savings by room type, office zones show the lowest energy savings and shops demonstrate the most significant energy-saving potential. As for shops, there are a wide range of unit capacities, from 4 to 25.7 kW, which contributes to substantial variations in energy-saving potential. Additionally, it was demonstrated that higher capacity units achieve greater energy savings with optimal start methods.

Table 5 summarizes the energy savings with the optimal start control in the heating mode for Building B. Fig. 16 shows the average energy savings per day categorized by zone type and HP power. The office zones show the lowest energy savings and shops demonstrate the highest energy-saving potential. Both offices and shops show variations in energy-saving potential due to the range of unit capacities. In general, the optimal start control results in greater energy savings for units with higher capacity.

4.2. Evaluation of optimal start control performance for new field test data

To evaluate the four optimal start control algorithms in real buildings, they were deployed in the field in two office buildings located in Eastern Washington; however, results from only one building are presented. The results from the second building are similar. The first real building (referred to here as Building A has 11 RTUs. The optimal start controls were run over a test period of 4 weeks during the heating season (March/April) and cooling season (July/August). The tests were monitored in real time and all trend data over the test period were stored at a 1-minute sampling interval. As described previously, each of the four optimal start models is trained using the last 10 weekdays of rolling data. The model parameters are updated once every 24 h after the actual optimal start time has been measured. The runtime for optimal start was configured to vary between 120 and 10 min, i.e., the maximum runtime is constrained to 120 min and the minimum runtime to 10 min. Many times, the zone temperature during the unoccupied period is close to the occupied set point; when this occurs, the RTU is run for at least 10 min before the start of occupancy.

Fig. 17 and Fig. 18 show time series data from a prototypical day from the experimental test results during the heating and cooling seasons. The results of all the experimental tests show the zone temperature (left axis) and compressor signal and outdoor air temperature (right axis) on the dual y-axis. The zone is shown where the algorithm changes the set points from unoccupied to occupied values at different times. In each case, the algorithm changes the set point at a time that is predicted to enable the zone temperature to reach the new value by 4:40 a.m. The RTU is activated based on the optimal start and in this case starts at the earliest time. The optimal start algorithm waits as long as possible before starting RTUs to minimize runtime while ensuring occupant comfort.

The predictions of Model 1 were worse than those of Model 2 when dealing with large increases in zone temperature during optimal start (difference between zone temperature and zone set point). When the increments were bigger, however, Model 1 behaved better. The

predictions of Model 3 showed slightly better accuracy compared to the other models (Models 1, 2, and 4), with the optimal start time predicted, on average, to be less than 7 min when considering factors such as the time constant of the zone. The results showed a minimal influence of outside temperature on optimal start time estimation, as the parameter dependent on the outside temperature was always a very low value. Therefore, when forecasting the preheat time, the main parameter to be considered should be room temperature, which, to some extent, reflects the effect of the outside temperature. Only in very special circumstances should the influence of outside temperature be included in the controller. These results illustrate the ability of the algorithm to accurately predict the start time to meet the desired set point at the occupancy time.

5. Conclusion

Because optimal start controls can result in significant energy savings, four data-driven adaptive optimal start algorithms that can be deployed in SMBs with RTUs were developed. While the models have similarities in assessing deviations between zone temperature and set point temperature, they differ in their consideration of outdoor temperature and start time adjustments. These models demonstrated good performance in terms of reduced operation time and temperature control, with a mean temperature-based accuracy error of less than 0.5 °C and a mean time-based error of less than 15 min.

These methods have been shown to result in significant energy savings (2 to 5 kWh/day/unit) and reduction in pre-occupancy runtime of the units. Energy-saving potential varied based on several factors, including building layout, room type, season, and the unit rated power capacity. The units on the east side of the building showed higher savings, while interior units showed greater variability in savings due to the differences in RTU capacities and room sizes. Saving was considerably greater during the heating season compared to the cooling season.

All models performed well in field tests, with Model 3 exhibiting slightly superior accuracy. Moreover, the models exhibited minimal sensitivity to outside temperature influences in optimal start estimation. The results obtained have also shown that when using adaptive optimal start models, the outdoor temperature measurement could be neglected when estimating the correct precooling or preheating time. The main parameter to be considered is the zone temperature, which to some extent reflects the effect of the outdoor temperature. Thus, the optimal start model could be simplified to have only two parameters.

High-quality real building data is crucial for accurate parameter estimation and the capability of the models. The accuracy of the models increases with an increase in the training data. After 10 days the parameters of the models tend to be accurate, even though the initial estimates are somewhat inaccurate. The models generally achieve good performance with just 1 to 2 weeks of training data. The minimal sensor requirement (zone temperature and outdoor air temperature) is the unique feature of the algorithm.

A consistent issue was found on Mondays after the zone temperature set point was setback for the duration of the weekend. The zone's thermal mass has stored or released (for cooling and heating respectively) significant heat energy that must be recovered during the optimal start period. This resulted in a considerable difference in building thermal mass response and higher inaccuracy in the predicted optimal start time. In the future, it may be advantageous to have a separate model for Mondays and the day after a holiday to reduce the prediction error. It is also recommended that additional research and development be carried out to implement the optimal start control for systems that incorporate multiple evaporators and variable-speed compressors. As systems with multiple evaporators have been increasing in the world market, tests extending the data-driven optimal start algorithms to these systems are recommended. When the optimal start algorithm is extended to systems with multiple evaporators, some modifications and adjustments should be required for accurate estimation. Further

performance evaluations are also recommended for systems with variable-speed compressors.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Srinivas Katipamula and all co-authors reports financial support was provided by United States Department of Energy Building Technologies Office. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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Appendix

This appendix presents the energy-saving analysis of the optimal start method.

Figs. 19 and 20 show the average optimal start runtime (histogram) for all units in Building B during summer and winter seasons, respectively. These figures also provide a comparison with the average runtime to setpoint and total runtime with a fixed schedule.

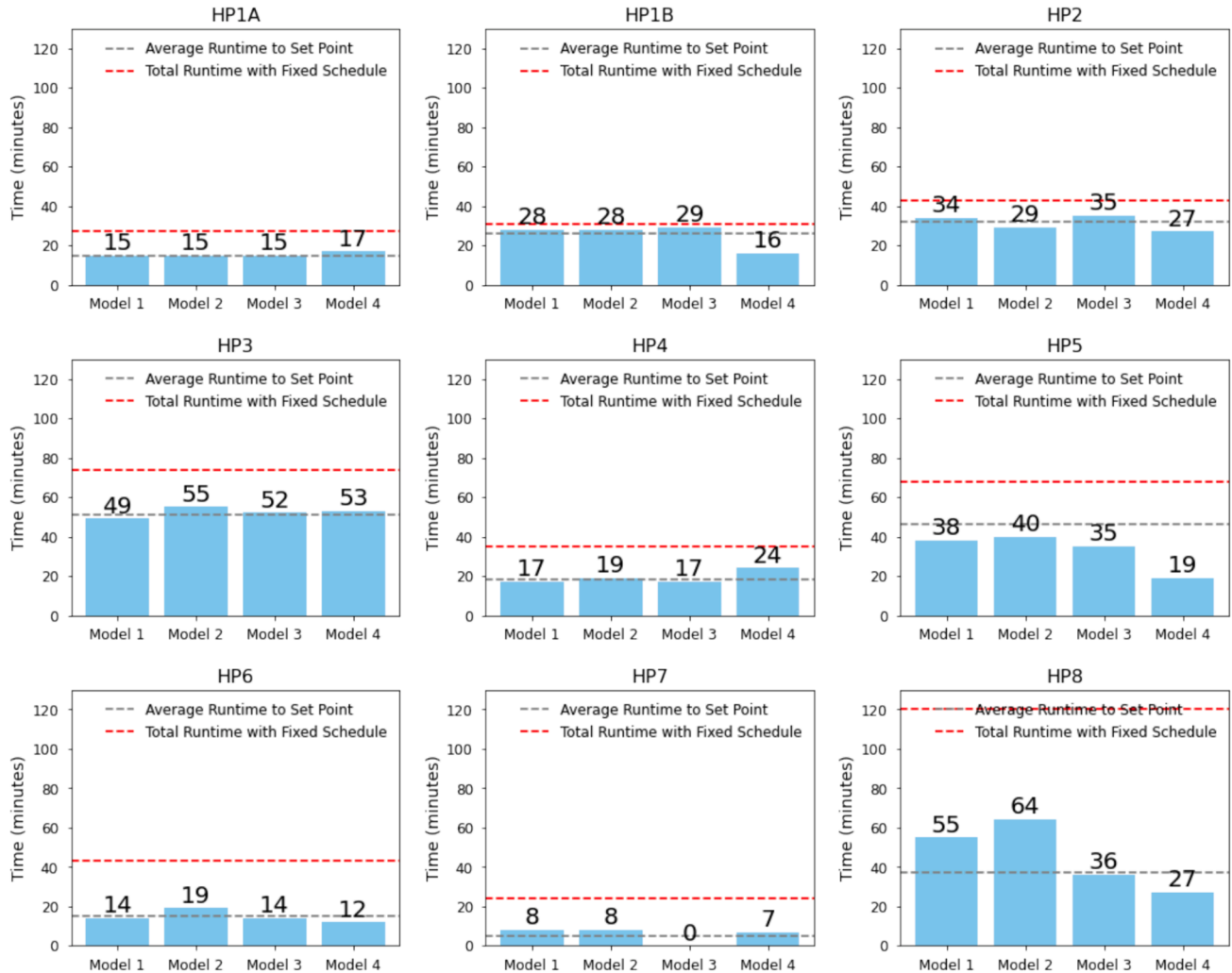


Fig. 19. Average optimal start control runtime of all units in Building B during summer (July–August).

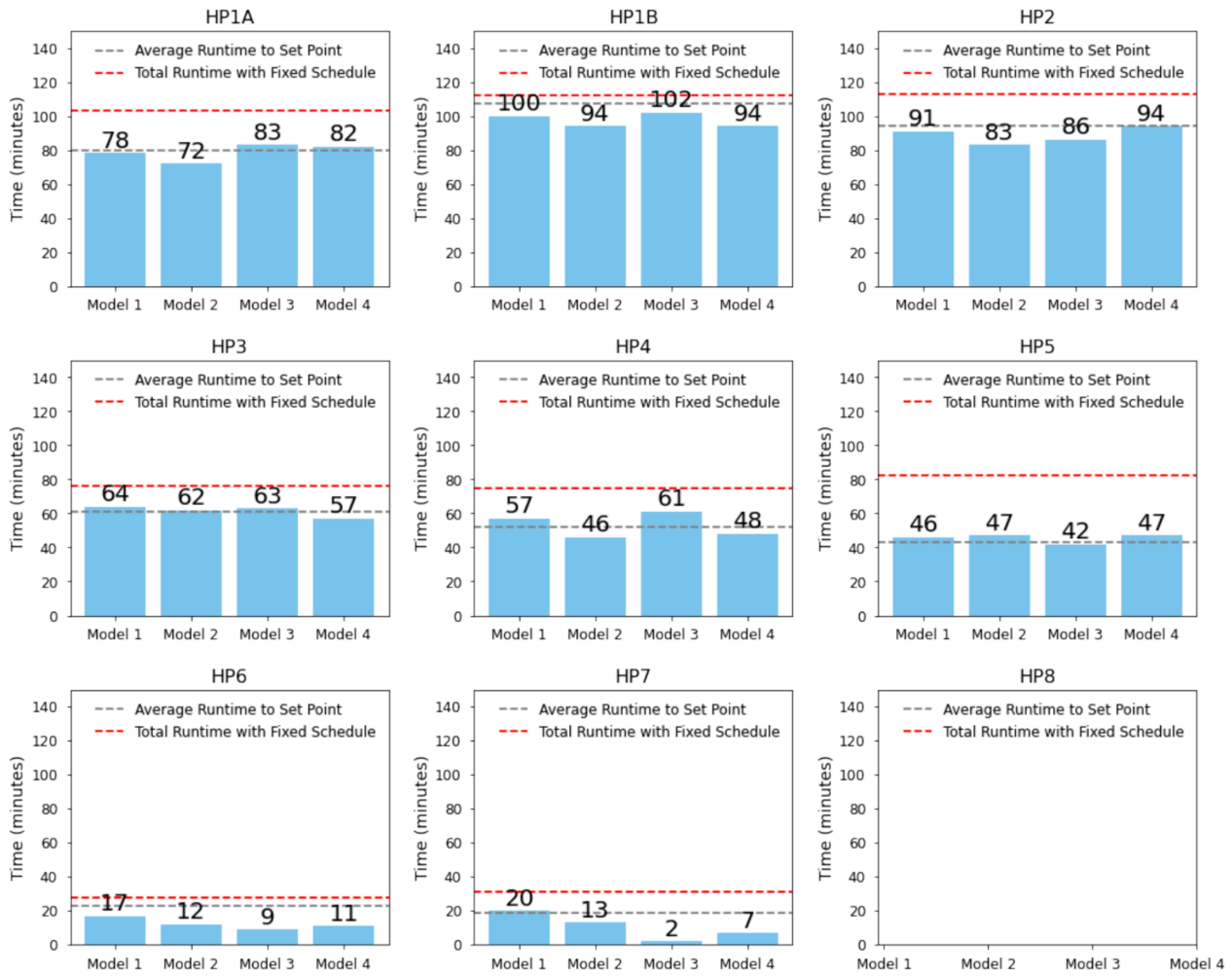


Fig. 20. Average optimal start control runtime of all units in Building B during winter (November– February).

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